HW1: Particle System for Soft-body Simulation

Computer Animation and Special Effects, Spring 2026

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1. Implementation

1.1 Spring Force and Damper Force

The internal forces of the jelly are composed of spring forces and damper forces acting along each spring connection. The spring force follows Hooke's law: $\mathbf{f_a = -k_s(|x_a - x_b| - r) * l}$, where **l** is the unit vector from particle B to A, $\mathbf{k_s}$ is the spring coefficient, and **r** is the rest length. When the distance exceeds the rest length, the force pulls the particles together; when shorter, it pushes them apart.

The damper force is: $\mathbf{f_a = -k_d((v_a - v_b) \cdot l) * l}$, where $\mathbf{k_d}$ is the damping coefficient. It acts along the spring direction and resists relative motion, causing the system to gradually lose energy and come to rest.

In *computeInternalForce()*, all springs are iterated. For each spring, the spring force and damper force are computed and applied symmetrically to both endpoint particles (Newton's third law: +F on A, -F on B).

1.2 Collision Detection and Response

Collision with the ground plane is handled in *handleCollision()*. The plane has a normal vector **N = (sin(20°), cos(20°), 0)** and passes through the origin. Two cases are handled:

Case 1 — Collision: When a particle is close to the plane ($N \cdot (x - p) < \varepsilon$) and moving toward it ($N \cdot v < 0$), the velocity is decomposed into normal and tangential components. The normal component is reflected with a restitution coefficient: $\mathbf{v' = -k_r V_N + V_T}$, where $k_r = 0.8$.

Case 2 — Contact: When a particle rests on the surface ($|N \cdot (x-p)| < \varepsilon$, $|N \cdot v| < \varepsilon$, and $N \cdot f < 0$), a contact force $\mathbf{f^c = -(N \cdot f) N}$ is applied to resist penetration, along with a friction force $\mathbf{f^f = -k_f(-N \cdot f) v_t}$ to decelerate tangential motion.

Finite Plane Range: The ground plane spans only 30×30 units. To account for this, each particle's position is projected onto the plane's local coordinate system before collision processing. The local X-axis is computed as the direction perpendicular to the normal within the XY plane: **planeX = (cos(20°), -sin(20°), 0)**, and the local Z-axis remains **(0, 0,**

1). If the projected coordinates exceed ±15 units (half of 30) in either axis, the particle is outside the platform and collision is skipped. This allows the jelly to slide off the edge of the slope and fall freely, which is physically correct and avoids artificial collision with a nonexistent infinite plane.

1.3 Integrators

Explicit Euler: The simplest method. Position is updated using the current velocity, then velocity is updated using the current acceleration: $\mathbf{x_{n+1} = x_n + v_n \cdot dt}$, $\mathbf{v_{n+1} = v_n + a_n \cdot dt}$. It is straightforward but can be unstable for stiff systems with large time steps.

Implicit (Semi-Implicit) Euler: Velocity is updated first using the current acceleration, then position is updated using the *new* velocity: $\mathbf{v_{n+1} = v_n + a_n \cdot dt}$, $\mathbf{x_{n+1} = x_n + v_{n+1} \cdot dt}$. This symplectic Euler approach provides better energy conservation and stability compared to explicit Euler, as the position update uses information from the future time step (the updated velocity).

Runge-Kutta 4th Order (RK4): A high-accuracy method that evaluates the derivative at four points within each time step: k1 at t, k2 and k3 at t+dt/2, and k4 at t+dt. The final update uses a weighted average: $\mathbf{x_{n+1} = x_n + (dt/6)(k1 + 2k2 + 2k3 + k4)}$. At each intermediate step, *computeJellyForce()* is called to recompute forces at the predicted positions. RK4 provides 4th-order accuracy ($O(h^5)$ local error) and handles the stiff spring system well.

Position-Based Dynamics (PBD): Unlike force-based methods, PBD operates on positions directly. The pipeline consists of three steps:

(1) **Prediction:** Positions are predicted using explicit Euler and stored as predicted positions. (2) **Constraint Projection:** Iteratively enforce terrain constraints (push particles above the ground plane) and distance constraints (keep neighboring particles within min/max distance bounds). (3) **Velocity Update with XSPH:** New velocity is derived from position change (predicted - original)/dt, then smoothed using XSPH viscosity, which blends each particle's velocity with its neighbors' velocities using inverse-distance weighting.

2. Result and Discussion

2.1 Difference Between Integrators

Integrator	Stability	Accuracy	Visual Behavior
Explicit Euler	Low (h < 2/k required)	1st order	Jelly bounces naturally; can explode if dt too large
Semi-Implicit Euler	Medium (better energy conservation)	1st order	Similar to Explicit Euler; slightly more stable at contact

RK4	High (large stability region)	4th order	Smoothest motion; best bounce quality
PBD	Very High (unconditionally stable)	N/A (not ODE-based)	Rigid appearance; no spring oscillation

Explicit Euler is the simplest but least stable. For the default parameters (springCoef=2000, damperCoef=60, dt=0.001), it performs adequately. RK4 provides the highest accuracy with smooth, natural-looking motion. PBD is unconditionally stable but produces a visually rigid jelly since it constrains positions directly rather than using spring forces. The semi-implicit Euler offers a good balance between simplicity and stability.

2.2 Effect of Parameters

Spring Coefficient (k_s): Default = 2000. Higher values make the jelly stiffer and less deformable. Lower values produce a softer, more elastic jelly. Very high spring coefficients can cause instability in Explicit Euler if dt is not reduced accordingly (h < 2/k constraint).

Damper Coefficient (k_d): Default = 60. Controls how quickly oscillations decay. Higher damping makes the jelly settle faster but appear less elastic. Lower damping allows prolonged bouncing. Zero damping results in perpetual oscillation.

Coefficient of Restitution (k_r): Default = 0.8. Controls energy retained after collision. 1.0 = perfectly elastic bounce (no energy loss). 0.0 = perfectly inelastic (no bounce). 0.8 retains 80% of normal velocity after impact.

Coefficient of Friction (k_f): Default = 0.5. Controls resistance to sliding along the surface during contact. Higher values cause the jelly to decelerate faster on the slope. Zero friction allows free sliding.

Delta Time (dt): Default = 0.001. Smaller dt improves stability and accuracy but increases computation time. For Explicit Euler with k=2000, dt must satisfy h < 2/k = 0.001 for convergence. RK4 can tolerate larger dt due to its higher-order accuracy.

PBD-specific parameters: Solver iterations (default=4) control constraint stiffness; more iterations = harder jelly. XSPH factor (default=0.1) controls velocity smoothing; higher = more viscous behavior. Min/max distance ratios (0.75/1.25 of cell size) define the allowable deformation range for distance constraints.

3. Bonus: Runge-Kutta 4th Order

The RK4 integrator was implemented as a bonus. It evaluates the system's derivative (velocity, acceleration) at four points per time step, using *computeJellyForce()* to recompute forces at each intermediate state. The four stages are:

k1: Evaluate at the original state (t_n).
k2: Evaluate at t_n + dt/2, using k1 to predict the intermediate state.
k3: Evaluate at t_n + dt/2, using k2 to predict the intermediate state.
k4: Evaluate at t_n + dt, using k3 to predict the state.

The final update combines all four evaluations with weights 1:2:2:1, yielding 4th-order accuracy. Compared to Explicit Euler, RK4 produces noticeably smoother trajectories and better handles the stiff spring system. The trade-off is that it requires 4x the force evaluations per time step, making it approximately 4 times slower. However, this higher cost is justified by the significantly improved accuracy and larger effective stability region.

4. Conclusion

This assignment implemented a soft-body jelly simulation using a mass-spring system with four different integration methods. Key findings include:

(1) **Explicit Euler** is simple and fast but requires small time steps for stability. (2) **Semi-Implicit Euler** provides better stability with minimal additional complexity by updating velocity before position. (3) **RK4** achieves the best accuracy and smoothest visual results at the cost of 4x computation. (4) **PBD** is unconditionally stable and suitable for real-time applications, but produces a visually stiffer jelly since it operates on positions directly rather than through force-based dynamics.

The collision handling with velocity reflection and contact forces (resist + friction) works correctly across all integrators. The inclined plane (-20 degrees around Z-axis) provides a good test case for observing sliding, bouncing, and friction effects. The spring coefficient and damper coefficient significantly affect the jelly's behavior, and proper tuning is essential for achieving realistic soft-body motion.